Fiscal Year:	FY 2022	**Task Last Updated:**	FY 12/09/2021
PI Name:	Haveman, Natasha Ph.D.		
Project Title:	Plant Microbiome - Its Influence in Plant Growth and Protection Against Pathogenesis in Spaceflight		
Division Name:	Space Biology		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Microbiology (2) Plant Biology		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	(1) Bioregenerative Life Support		
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Zip Code:	32953	**Congressional District:**	8
Comments:	NOTE: Dr. Natasha Haveman, formerly at the University of Florida (UF), joined the NASA Kennedy Space Center (KSC) Space Biology team in 2023. NOTE: name change to Haveman née Sng (Ed., June 2021)		
Project Type:	Ground	**Solicitation / Funding Source:**	2016-17 Space Biology (ROSBio) NNH16ZTT001N-FG. App G: Flight and Ground Space Biology Research
Start Date:	02/13/2019	**End Date:**	02/12/2022
No. of Post Docs:	1	**No. of PhD Degrees:**	1
No. of PhD Candidates:		**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:		**Monitoring Center:**	NASA KSC
Contact Monitor:	Freeland, Denise	**Contact Phone:**	321-867-5878
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Flight Program:			
Flight Assignment:	NOTE: End date changed to 2/12/2022 per NSSC information (Ed., 5/25/21)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Foster, Jamie Ph.D. (MENTOR: University of Florida, Gainesville)		
Grant/Contract No.:	80NSSC19K0520		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	A. Project Description - Long-duration exploration in space has emphasized the importance of understanding how to grow plants under microgravity, so as to supply crews with a continuous source of fresh food as well as provide psychological benefits. Plant and animal microbiomes have been known to play a significant role in maintaining host health and development through the regulation of metabolic and immune responses. Therefore, understanding the microbial population associated with plant growth and suppressing multi-host pathogenesis in spaceflight is not only crucial for sustaining the production of edible crops in space but also vital for the safety of crew members. The objective of this ground-based investigation is to develop a refined understanding of the basal plant microbiome of salad crops grown in plant growth chambers under previously used International Space Station (ISS) growth conditions and assess its potential for plant growth enhancements and human pathogenicity. Defining the basal plant microbiome will enable us to understand how spaceflight causes alterations in plant-microbial interactions and how to utilize it for future space exploration efforts. Hence, the overall goal of this project is to determine the influence of microorganisms associated with salad crops in plant growth and protection against pathogenesis under conditions similar to the ISS by utilizing metagenomics and metatranscriptomic approaches. Our central hypothesis is that changes in gene expression within the microbial population are key to understanding whether the plant microbiome can promote the growth of these crops through growth enhancement pathways. To investigate this hypothesis the following objectives were proposed: Objective 1 – to define the metatranscriptome of three different crops grown in plant growth chambers under ISS conditions. Objective 2 – to develop an understanding of the metabolic pathways and signaling leading to plant growth enhancements and/or to plant/human pathogenesis identified from the metatranscriptomic data. Objective 3 – to develop methods for enhancing plant growth or suppressing the pathogenic activity without eradicating commensal or saprophytic microbes. B. Significance - The rationale behind this proposal is to advance the technical readiness necessary to support a sustainable source of fresh food production for extraterrestrial exploration, and this will be accomplished by establishing a strong foundation in understanding how plant-microbe interactions impact the safe and successful production of plants. At the completion of these studies, it is expected that information gathered on the basal microbiome of salad crops cultivated in growth chambers similar to environmental conditions on the ISS will identify those microbes that have the potential to result in multi-host pathogenesis. The proposed metatranscriptomic studies, together with the metagenomic studies conducted by the space food production team at Kennedy Space Center (KSC), will help elucidate the functional pathways and mechanisms by which microbes interact with their host under ISS growth conditions. These data will then allow us to make crucial and informed engineering choices on plant growth hardware systems and crop selection for future space exploration vehicles and habitats. Furthermore, this study will provide new insights into standard terrestrial plant biology as little is currently known about the microbiome of salad crops.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	There exists a limited amount of knowledge about how the vast majority of plant-associated microorganisms impacts the plant's growth, health, and susceptibility to diseases. To date, the interplay between plants and microorganisms has been largely studied for various leaf pathogens, symbiotic rhizobia, and mycorrhizal fungi. However, little is known about the microbiome of salad crops. Salad crops require minimal processing before consumption and are used to investigate the baseline microbial data for developing space-related produce sanitation methods for safe consumption by the crew. So far, microbial studies performed on salad vegetables, such as spinach, lettuce, and cilantro, are mostly culture-dependent methods to determine whether human pathogens survive and persist on the edible surface of the leaves. A survey of the salad microbiome and how it plays a role in maintaining the host's health and development is lacking. Investigations conducted in spaceflight have revealed that microgravity does alter the plant-microbe interaction at both the genetic and physiological level, with a tendency to increase the susceptibility to plant pathogenesis. It is therefore necessary that a cultivation-independent analysis of the microbial communities found on salad crops be investigated with multi-omics, predominantly metagenomics and metatranscriptomics.
	Task Progress for Years 1-3: In total, we have two published manuscripts that stem from this postdoctoral fellowship, three oral presentations for the American Society for Gravitational and Space Research (ASGSR) (2020-2021), one poster presentation at ASGSR (2019), and another manuscript in preparation. [Ed. Note: for publications, see Cumulative Bibliography. For presentations, see list at end of Task Progress section.] Research Progress for each Objective: Objective 1 – to define the metatranscriptome of three different crops grown in plant growth chambers under ISS conditions. Major Outcomes: The major outcomes of this objective were: 1) The 28-day old lettuce metatranscriptome was established and published using the MinION sequencing platform. 2) Illumina NOVAseq sequencing of all lettuce samples at 4 different stages of development was completed and data analysis is nearly completed. 3) High-quality RNA has been extracted from Mizuna samples and optimization for library sequencing has been completed. Details of work progress: In YR1 of the project, the "veggie team" at Kennedy Space Center (KSC) completed the growth of all three crops (lettuce, tomato, and mizuna). However, the tomato samples were harvested by the KSC veggie team in a different manner (liquid nitrogen) compared to the lettuce and mizuna samples (RNALater). Hence, the tomato samples were not used for further investigation in this proposal. In YR1-2: To ensure high-quality metatranscriptomic analysis, a pilot sample of lettuce was used to optimize a protocol for RNA extraction, sample processing, sequencing, bioinformatics pipeline, and metatranscriptomic analysis. Here, a sequencing experiment was done on a portion of the Day 28 lettuce leaf samples. In this experiment, the Oxford Nanopore MinION sequencer (MinION) was used to analyze the lettuce leaf microbiome. MinION sequencing is the targeted sequencing platform NASA plans to use for future spaceflight sequencing-related experiments. Thus, using the MinION device to sequence the 28-day lettuce leaf microbiome not only enabled us to optimize a protocol for using the MinION to perform quality control surveillance on future space food production; it also enabled us to optimize a bioinformatic pipeline for metatranscriptomic analyses. In addition, further validation can be done when comparing the

Task Progress:

gold standard Ilumina NOVAseq sequencing method with the new MinION sequencing technology.

In YR3 of the project, a thorough analysis of the pilot 28-day old lettuce leaf metatranscriptome (performed on the MinION sequencing platform) was completed and a paper was published. [Ed. Note. Reference: Haveman NJ, Khodadad CL, Dixit AR, Louyakis AS, Massa GD, Venkateswaran K, Foster JS. "Evaluating the lettuce metatranscriptome with MinION sequencing for future spaceflight food production applications." npg Microgravity . 2021 Jun;7:22. https://

In parallel, the full set of 24 lettuce samples (sanitized root and leaf samples at 7, 14, 21, and 28 days old) were sequenced via the Illumina NOVAseq 6000 platform. Sequencing was performed at the University of Florida, Interdisciplinary Center for Biotechnology Research (ICBR), and has been completed. The data obtained were run through an optimized bioinformatics pipeline (completed in YR2) and preliminary data for this study were obtained in YR2 as well. The data obtained show that root and leaf samples vary in their differential gene expression profiles and should be analyzed separately. A preliminary analysis of the lettuce leaf and root microbiome at various developmental time points has been established. Further pairwise comparisons between the abundance levels of the microbial communities for each developmental age are currently ongoing.

Further analysis of the transcriptome was performed in YR3 on this dataset and the differentially expressed genes (DEG) data have been gathered for each of the comparisons. In the data gathered, many more differentially expressed genes were observed in the root samples than in the leaf samples. Results show that the majority of statistically significant DEGs (padj < 0.05) in the roots can be seen in comparisons with Day 28, and therefore suggests that Day 28 has a substantially different metabolic and functional profile than the other developmental time points. In the leaf samples, the same trend was not observed. Although Day14 vs Day 28 showed the highest number of differentially expressed genes, other pairwise comparisons with Day 28 samples did not consistently show a substantial number of DEGs. Further comparisons of the individual genes are still ongoing.

In addition, a pilot study to optimize the sample processing method for the mizuna plants is ongoing. Findings from YR1 have shown that crop-specific ribosomal probes will need to be designed for the various host organisms to improve the removal of host RNAs and facilitate the microbial RNA enrichment process. In YR2, RNA from 28-day old mizuna leaf tissues have been extracted and several different commercially available ribosomal probes were tested. These samples have been shipped to our collaborator at Cornell University for sequencing, and data obtained will be used to validate the most effective method for enriching the mizuna microbial RNA. Due to COVID-19 and the shortage of laboratory supplies, our mizuna samples were not processed for quite a while. At the end of YR3, the sequencing of these samples for the pilot samples has been completed and data is currently being transferred to the Principal Investigator (PI) for further downstream bioinformatic analyses. The results gathered will identify the best method of removing plant ribosomal probes for future RNAseq library construction.

Objective 2 – to develop an understanding of the metabolic pathways and signaling leading to plant growth enhancements and/or to plant/human pathogenesis identified from the metatranscriptomic data.

Major Outcomes:

The major outcomes of this objective were: 1) The metabolic pathways of the 28-day old lettuce leaf microbiome using MinION sequencing technology have been published. 2) Presence of potential bacterial genera and metabolic activity that could play a role as plant growth enhancers and plant pathogens were revealed. 3) Gene network analysis identified several genes within the fungal Mitogen-Activated Protein Kinase (MAPK) pathway that could be drivers of fungal pathogenesis. 4) The metabolic pathways of the lettuce leaf microbiome at all the various time points have been classified.

Details of work progress: In YR2, we have established the microbial populations found in the 28-day old lettuce leaves and the metabolic pathways that these microbes are utilizing. [Ed. Note. Reference: Haveman NJ, Khodadad CL, Dixit AR, Louyakis AS, Massa GD, Venkateswaran K, Foster JS. "Evaluating the lettuce metatranscriptome with MinION sequencing for future spaceflight food production applications." npg Microgravity. 2021 Jun;7:22. https://

In YR3, a similar approach was taken to evaluate the metabolic pathways for all the lettuce samples at various time points in their development. By looking at the pairwise comparison of differentially expressed genes for both the bacterial and fungal community, a sense of the pathways changing at each developmental time point can be inferred. The results of the changes in functional pathways of the leaf-bacterial community show that there are many diverse pathways differentially regulated between the 14-day old plants and the 28-day old plants. Genes associated with the metabolism of amino acids, carbohydrates, energy, vitamins and cofactors, replication and repair, and signal transduction are differentially expressed between these two developmental time points. Similar analyses, between the bacterial community in the root tissues, are currently ongoing.

Objective 3 – to develop methods for enhancing plant growth or suppressing the pathogenic activity without eradicating commensal or saprophytic microbes.

The results obtained from the 28-day old lettuce leaf microbial metatranscriptome analysis (with the MinION platform) have led to a potential strategy for the suppression of fungal plant pathogens. The taxonomy diversity data revealed the presence of several potential plant growth-promoting bacteria (such as Paenibacillus) and the presence of fungal plant pathogens (such as Fusarium). Several studies have shown that Paenibacillus produces an antifungal compound that is active against Fusarium. So perhaps, one way to combat the Fusarium presence is to enhance the presence of Paenibacillus in the population.

Other Projects Developing from this Grant:

In addition to the proposed project for this fellowship, an additional paper on using the MinION platform as a potential diagnostic tool for the detection of an opportunistic fungal pathogen identified on the ISS was developed. [Ed. Note: see bibliography for pre-print information and upcoming print publication date.] In collaboration with Dr. Andrew Schuerger at the Space Life Sciences Lab (SLSL) at Kennedy Space Center, a side experiment showing how the MinION platform can be used to accurately differentiate between Fusaria species was performed. The results show that the pipeline could differentiate between the kingdoms and species, and hence, strengthen the case for using the MinION platform as a rapid plant disease diagnostic tool in space.

	Haveman NJ and Schuerger AC. (2021) Diagnosing an Opportunistic Fungal Pathogen on Spaceflight Grown Plants Using the MinION Sequencing Platform. Astrobiology (In Press). Preliminary results obtained from this side project were published and used to secure another NASA Space Biology grant: Early Career Investigation-80NSSC22K0160: 1/01/2022 – 12/31/2022; $100,000 (PI: Haveman) Developing A System for Rapid Diagnosis of Plant Diseases and Monitoring of Plant Microbiome for Spaceflight Applications. Manuscripts: Haveman NJ, Khodadad C, Dixit A, Massa G, Louyakis A, Venkateswaran K, Foster JS. (2021) Evaluating the lettuce metatranscriptome with MinION sequencing for future spaceflight food production applications. Nature microgravity 7(1):1-11. Haveman NJ and Schuerger AC. (2021) Diagnosing an Opportunistic Fungal Pathogen on Spaceflight Grown Plants Using the MinION Sequencing Platform. Astrobiology (In Press). Haveman NJ, Khodadad C, Dixit A, Massa G, Venkateswaran K, Foster JS. Comparative Metatranscriptomics of the lettuce microbiome reveals organ-specific mutualisms. (In preparation). Oral Presentations: 2020 American Society for Gravitational and Space Research (ASGSR), Virtual (Technical session: Plants). Sng NJ, Khodadad C, Dixit A, Massa G, Louyakis A, Venkateswaran K, Foster JS. Plant-Microbial Metatranscriptomic Using Nanopore Sequencing Platform for Spaceflight Applications. 2021 American Society for Gravitational and Space Research (ASGSR), Baltimore, MD. Haveman NJ. Space Biology Program Session - Early Career Lighting Talks. Using the MinION for Plant Growth Studies in Space. 2021 American Society for Gravitational and Space Research (ASGSR), Baltimore, MD. Haveman NJ and Schuerger AC. Diagnosing an Opportunistic Fungal Pathogen on Spaceflight-Grown Plants Using the MinION Sequencing Platform. (Technical session: Plants) Poster Presentations: Sng NJ, Foster JS, Venkateswaran K, Massa G, Khodadad C, Dixit A. Plant-Microbial Metatranscriptomic Analysis for Spaceflight Application. American Society for Gravitational Space Research, Denver, CO. November 2019.
Bibliography Type:	Description: (Last Updated: 08/03/2023)
Articles in Peer-reviewed Journals	Haveman NJ and Schuerger AC. "Diagnosing an opportunistic fungal pathogen on spaceflight-grown plants using the MinION sequencing platform." Astrobiology. 2021 Nov 18; https://doi.org/10.1089/ast.2021.0049 ; PubMed PMID: 34793258 , Nov-2021